

computer networks

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I am writing this document as a passion project for myself. This document will be writing mostly from my memory with some help from LLM(s).

A switch, for the most part, handles layer two traffic. Switches use the Address Resolution Protocol(ARP) to determine where on a Local Area Network(LAN) a given device with a certain Media Access Control(MAC) address may be. The ARP sends what is known as broadcasts out of all of its ports and waits for a reply from the machine with a given MAC address. Switches can act as a physical segmentation of a LAN and they can implement something known as a VLAN to logically segment that same network. The ARP can create broadcast storms and other issues locally so there is another useful protocol that switches use known as spanning tree protocols. There are a variety of spanning tree protocols: STP, RSTP, MSTP, PSTP, ect. each with their own nuances and benefits. There are also a variety of security protocols switches implement because after all switches are like the first layer into a network making them a valuable asset for attacks to target.

Routers can do many more things than switches in my opinion. Routers will look at layer two information like a switch, but that really only gets you to/from the next hop. Routers will look at layer three information. When I say layer three, I am referring to the OSI model (IP layer). There are two types of IP: IPv4, IPv6 each with their own reasons for existing. Routers can maintain routing tables which are similar in some ways to the MAC address tables that are maintained by switches. Routing tables can be manually populated with static routes set up by an admin. Routers can also alter the routing table dynamically with routing protocols. There are many routing protocols with their own caveats: OSPF, EIGRP, IS-IS, BGP, RIP, ect. Routers can have some what of a security feature called Access Control Lists(ACLs). These lists can filter traffic a little bit by looking at different encapsulation layers.